

UNIVERSITY OF CAPE TOWN

ECO4102Z/ECO4122Z

Active Management (Treynor Black Model): Activity 9a (SOLUTIONS)

1. You are given the following information about an active portfolio, denoted A, the market portfolio, denoted M, and the risk-free asset, denoted R.

	Active Portfolio (A)	Market Portfolio (M)	Risk-free asset (R)
Expected Return $E(R)$	7.3%	9%	2% r_f
Variance σ^2	10%	9%	0
Beta β	0.88	1	0
Jensen's Alpha α	0.02	0	0

Using the information given, calculate the optimal weight w_A^* of the *active portfolio* and interpret your answer.

2. You have recently been appointed as portfolio manager at TB Analytics. Your manager requires you to use the Treynor Black Model to advise a customer on his asset allocation decision. The customer has selected three securities for inclusion in his portfolio: FNB, BHP Billiton and Richemont. The regression statistics that you obtained for FNB are contained in Table 1, whilst Table 2 contains additional statistics that you may need. Use the information given to answer the questions that follow.

$$1. \quad S_p = \frac{7,3\% - 2\%}{\sqrt{0,1}} = 0,1676$$

$$0,1676^2 = 0 + \left(\frac{0,02}{\sigma(e_A)} \right)^2$$

$$\sigma(e_A) = 0,1193$$

$$\sigma^2(e_A) = 0,01424 \quad \times$$

$$w_A^0 = \frac{2\% / 0,01424}{(9\% - 2\%) / 9\%} = 1,8058 \quad \times 0,88$$

$$w_A^* = \frac{1,8058}{1 + (1 - 0,88) 1,8058} = 1,48 \quad 0,77$$

weight in active portfolio = 77% $\frac{1}{2}$ in mkt portfolio = 23%

$$\begin{aligned} \sigma_A^2 &= \beta_A^2 \sigma_m^2 + \sigma^2(e_A) \\ 10 &= (0,88^2 \times 9) + \sigma^2(e_A) \\ \sigma^2(e_A) &= 3,04 \end{aligned}$$

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- 30:00
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Expected Return	7.3%	9%	2%
Variance	10%	9%	0
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Jensen's Alpha α_A	0.02	0	0

Using the information given, calculate the optimal weight of the active portfolio ^{$\therefore$ Treynor} and interpret your answer.

- memo has error - α is not annualized
2. You have recently been appointed as portfolio manager at TB Analytics. Your manager requires you to use the Treynor Black Model to advise a customer on his asset allocation decision. The customer has selected three securities for inclusion in his portfolio: ^FFNB, ^BBHP Billiton and ^RRichemont. The regression statistics that you obtained for FNB are contained in Table 1, whilst Table 2 contains additional statistics that you may need. Use the information given to answer the questions that follow.

$$\sigma^2 = \text{Total risk}$$

$$\sigma^2(e_A) = \text{firm specific / unsystematic risk / tracking error}$$

F

Table 1: FNB CAPM regression results (calculated using monthly data)

Regression statistics				
Multiple R	0.8923			
R square R^2	0.5239			
Adjusted R square	0.5157			
Standard error	0.0767	std Err		
Observations	60	n		
ANOVA				
	Df	SS	MS	
Regression	1	0.3752	0.3752	
Residual	58	0.3410	0.0059	
Total	59	0.7162	$\sigma = \frac{SS_T}{n}$	
	Coefficients	Standard error	t-stat	p-value
Intercept	0.0086 α	0.0099	6.585	0.0000 sig
JSE ALSI	2.0348 β	0.2547	7.9888	0.0000 sig

Table 2: Additional information (annual statistics)

	JSE ALSI m	BHP B	Richemont R
SD σ	0.1358	0.2901	0.1935
Alpha α	0	0.0150	0.012
Beta β	1	1.23	0.62
SD of residuals $\sigma^2(\epsilon_i)$	0	0.2392	0.1757
Sharpe ratio of the complete portfolio = 0.4526 $= S_p$			

- a) Calculate the weightings of each of the three assets in the active portfolio. Provide possible reasons for the relative weightings obtained. (17)
- b) Calculate the weightings of each of the active and passive portfolios in the complete portfolio. (17)
- c) Use your answer in B) to calculate the risk premium and standard deviation of the complete portfolio. (6)

$$2a) \quad w_F^0 = \frac{0,0086}{0,0106} = 0,1218 \quad \checkmark$$

$$w_B^0 = \frac{0,0150}{0,2392} = 0,2622 \quad \checkmark$$

$$w_R^0 = \frac{0,012}{0,1757} = 0,3887 \quad \checkmark$$

$$w_F = \frac{0,1218}{\sum w_i^0} = 15,76\% \quad \checkmark$$

highest systematic risk $\checkmark$ + lowest α
 $\therefore$ lowest α/σ ratio

$$w_B = \frac{0,2622}{\sum w_i^0}$$

$$= 33,93\% \quad \checkmark$$

$$w_R = \frac{0,3887}{\sum w_i^0} = 50,3\% \quad \checkmark$$

lowest systematic risk + highest $\frac{\alpha}{\beta}$ risk

$$b) \quad \alpha_A = 15,76\% \times [(1+0,0086)^{12} - 1] + 33,93\% \times 0,0150 + 50,3\% \times 0,012 = 0,0125 \quad \checkmark \text{ meant to be annualized}$$

$$w_A^* \quad \sigma^2(e_A) = 15,76\%^2 \times 0,0106 + 33,93\%^2 \times 0,2392^2 + 50,3\%^2 \times 0,1757^2 = 0,0162$$

$$S_p^2 = S_m^2 + \frac{\alpha_A^2}{\sigma^2(e_A)}$$

$$0,4526^2 = \frac{RPM^2}{0,1388} + \frac{0,0125^2}{0,162}$$

$$RPM \approx 0,0605$$

$$w_A^* = \frac{0,0125 \div 0,0162}{0,0605 \div 0,1388} = 0,9804 \times 0,2352$$

$$\beta_A = \frac{w_i \beta_i}{1,05} \quad \checkmark$$

$$w_A^* = \frac{0,2352}{1 + (1-1,05)0,2352} = 0,238 \quad \checkmark \quad 23,8\% \text{ active}$$

$$w_m = 1 - 0,238 = 76,2\% \quad \checkmark \text{ passive (mkt)}$$

3. In your first assignment as a graduate, you have been handed the unfinished work of someone who recently left the company. His task was to utilize the Treynor Black method to derive an appropriate portfolio with the following shares: Sasol, RMB Holdings (RMBH) and Naspers. You now have to complete whatever is missing in order to report back to your client.

Use the information given to answer the questions that follow:

	ALSI <i>m</i>	Active Portfolio ^A	Sasol <i>S</i>	RMBH <i>R</i>	Naspers <i>N</i>	Complete portfolio ^P
$\sigma^2(e)$			0.0705	0.0392	0.0297	
$\alpha/\sigma^2(e)$		--	0.2128	0.1913	0.4040	
$w0_i$		1.0000	0.2633	0.2367	0.50	
α_A		--				
$\sigma^2(e_A)$		--				
$w0_A$		--				
W*(risky portfolio)	--	--				
Beta		β_A 1.1521	--	--	--	--
Risk premium	0.05 RP_m					--
SD	0.1552 σ_m	--				--

- a) You are given the following beta values, but you are not sure which beta value belongs to which company. Using the information given in the above table, identify which company is associated with each value, providing reasons for your answers.

Beta 1 = 2.0348, Beta 2 = 1.2672, Beta 3 = 0.6736 (5)

- b) Calculate the α_A and standard deviation of the active portfolio (9)
 c) Calculate the Expected return and Standard deviation of the Complete portfolio (11)

$E(R_p)$

σ_{CP}

(11)

a) $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i)$ Higher β = lower weight

$\therefore \beta_1 = \beta_R = 2,0348$ ✓ highest systematic risk

$\beta_2 = \beta_S = 1,2672$ ✓

$\beta_3 = \beta_N = 0,6736$ ✓ lowest systematic risk, highest weighting

b) $w_S^0 = 0,2128 = \frac{\alpha_S}{0,0705}$

$\alpha_S = 0,015$ ✓

$w_R^0 = 0,1913 = \frac{\alpha_R}{0,0392}$

$\alpha_R = 0,0075$ ✓

$w_N^0 = 0,4040 = \frac{\alpha_N}{0,0297}$

$\alpha_N = 0,012$ ✓

$\alpha_A = \frac{26,33\%}{21,28\%} \times 0,015 + \frac{23,67\%}{19,13\%} \times 0,0075 + \frac{56\%}{40,40\%} \times 0,012$
 $= 1,17\%$

$\sigma^2(e_A) = \sum w_i^2 \sigma^2(e_i)$
 $= 0,0145$ ✓

$\sigma_A = \sqrt{1,1521^2 \times 0,1552^2 + 0,0145}$
 $= 21,56\%$ ✓

c) $w_A^0 = \frac{1,17\% / 0,0145}{0,05 / 0,1552^2}$
 $= 0,3887$ ✓

$w_A^* = \frac{0,3887}{1 + (1 - 1,1521) \times 0,3887}$
 $= 0,4131$ ✓

$w_M^* = 1 - 0,4131$
 $= 0,5869$ ✓

$E(R_p) = (0,5869 + 0,4131 \times 1,1521) \times 0,05 + 0,4131 \times (1,17\%)$
 $= 5,8\%$ ✓

$S_p^2 = \frac{0,05}{0,1552} + \frac{1,17\%}{\sqrt{0,0145}}$
 $S_p = 0,6476$

$0,6476 = \frac{5,8\%}{\sigma_p}$

$\sigma_p = 0,0896$

$\sigma_p^2 = \frac{\sigma_m^2}{\beta_p^2} + \frac{w_A^*}{\beta_p^2} \times 0,0145$
 $\sigma_p^2 = \frac{0,1552^2}{1,1521^2} + \frac{0,4131^2}{1,1521^2} \times 0,0145$

$\sigma_p = 11,23\%$

